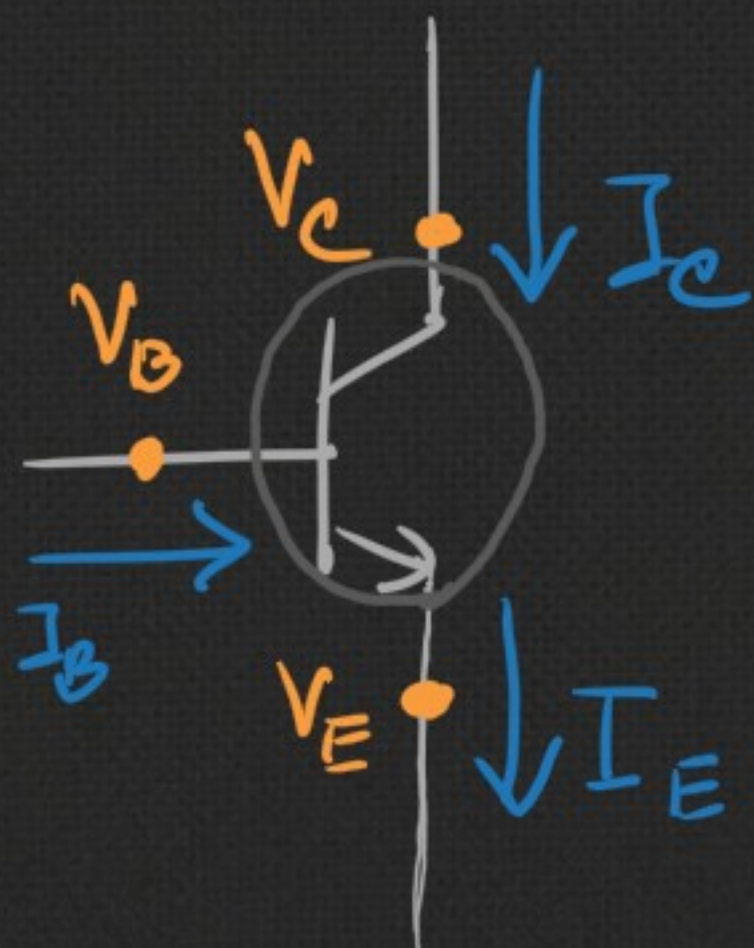


BJT in DC

KCL at supernode BCE $\Rightarrow$



$$I_E = I_C + I_B$$

$\rightarrow$ True for any mode.

In **ACTIVE**, $I_C = \beta I_B$ $\rightarrow$ ONLY in ACTIVE

$$\therefore I_E = I_C + \frac{I_C}{\beta} = \frac{1+\beta}{\beta} I_C$$

$$\therefore I_C = \frac{\beta}{1+\beta} I_E \rightarrow \text{ONLY in ACTIVE}$$

We define $\frac{\beta}{1+\beta}$ as α

$$\text{So, } I_C = \alpha I_E$$

The typical value for $\beta = 100$. So, $\alpha \approx 0.99$. So, we can say, during ACTIVE region, $I_C \approx I_E$.

In **SATURATION**, $V_{CE} = V_{CE(sat)} = 0.2V$. This is the lowest possible value of V_{CE} . This implies that the voltage V_{CE} in the CE configuration can be, at

$$\text{max, } \underline{V_{CC} - 0.2V}.$$

$\rightarrow$ That is, 0.2V less than supply



How to solve circuits with BJT?

⇒ Our old friend, Method of assumed state!

For BJT

① If assume in **ACTIVE** mode

$$\boxed{V_{BE} = 0.7, I_C = \beta I_B} \Rightarrow \text{Eqn true for ACTIVE}$$

You must show $V_{CE} > 0.2$

② If assume in **SATURATION**

$$\boxed{V_{BE} = 0.8, V_{CE} = 0.2} \Rightarrow \text{Eqn true for SAT.}$$

You must show $I_C < \beta I_B$,

that is $\frac{I_C}{I_B} < \beta$

③ If assume in **CUTOFF**, $\boxed{I_C = I_B = I_E = 0.}$ $\Rightarrow$ Eqn true for CUTOFF

You must show $V_{BE} \leq 0.7V$ and $V_{CE} > 0.2$

3 steps

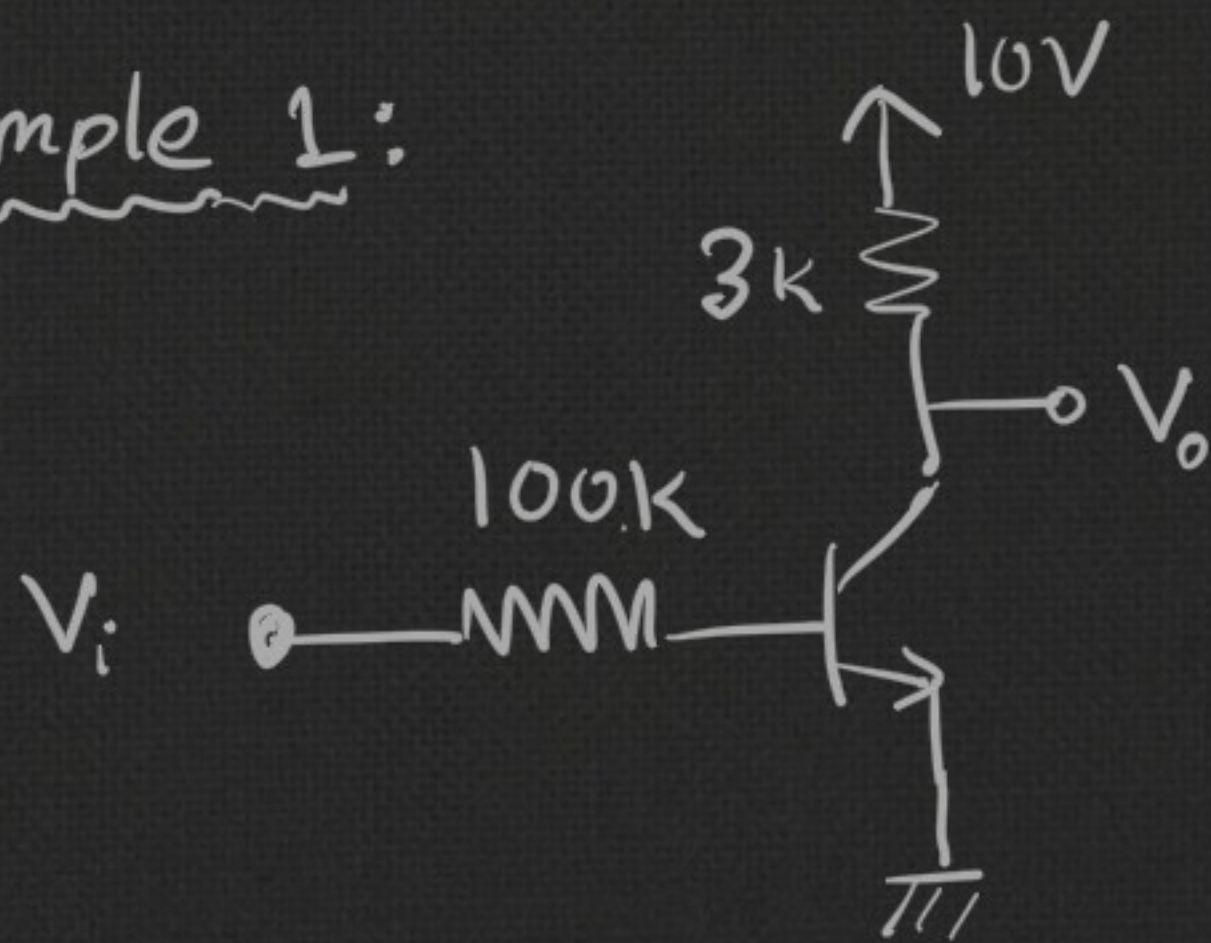
① Assume a state. there are 3 possible states: ACTIVE, SATURATION, CUTOFF

② Solve the ckt and find I_C, I_B, V_{CE}, V_{BE}

③ Verify your assumption. If wrong, repeat ①, ②, ③

*must show $\Rightarrow$ condition

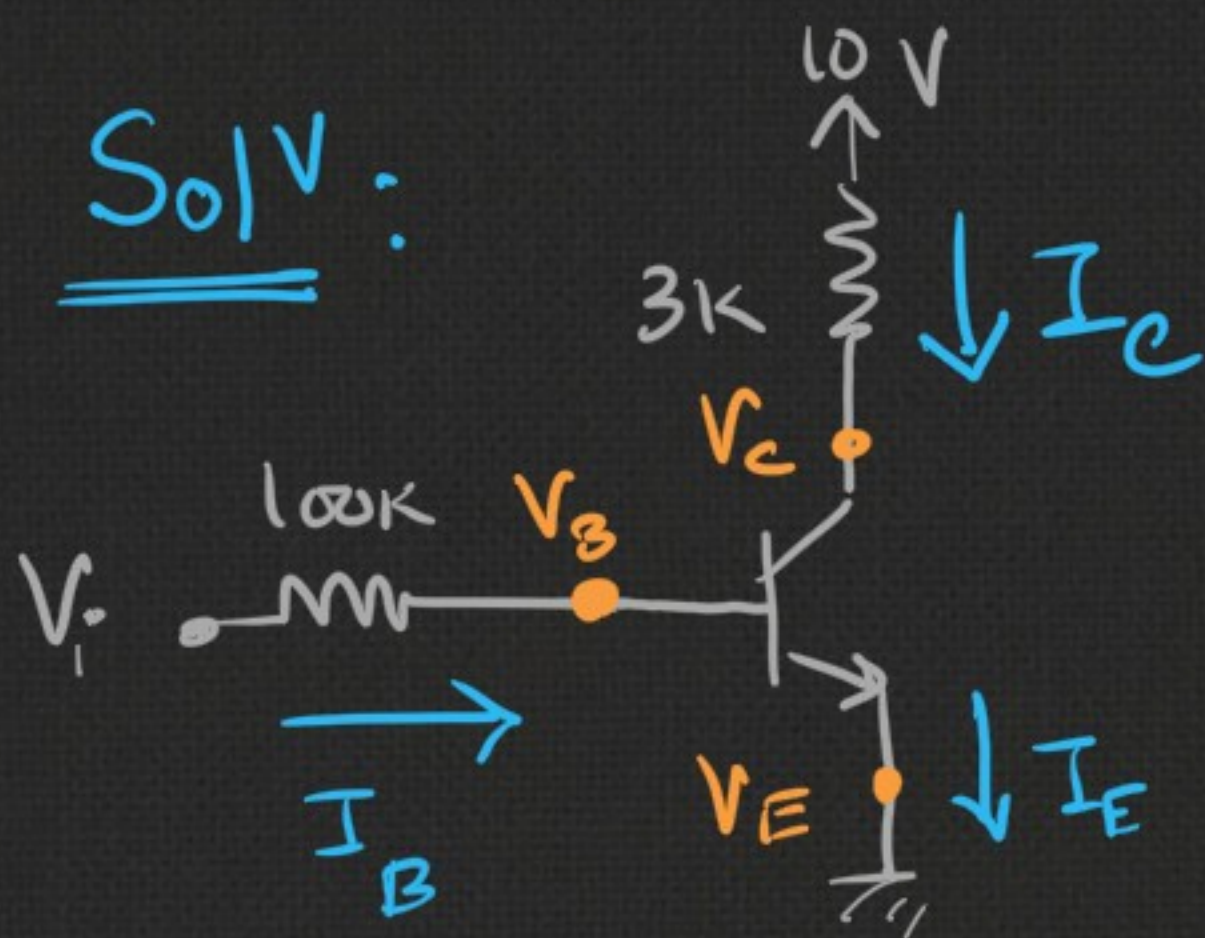
Example 1:



$$\underline{\underline{\beta = 100}}$$

Find V_o for (i) $V_i = 1V$ (ii) $V_i = 5V$

Solⁿ:



$$\boxed{V_o = V_c}$$

Since $V_i > 0.7V$ in both (a) and (b), it will be either in **ACTIVE** or **SATURATION**

$$BE = ON \Rightarrow V_{BE} = 0.7V$$

$$\text{Since } V_E = 0V \Rightarrow \therefore V_B - V_E = 0.7 =$$

$$\Rightarrow V_B = 0.7V$$

① Assume BJT in **ACTIVE**

$$\therefore I_B = \frac{V_i - V_B}{R_B} = \frac{1 - 0.7}{100} = 0.003 \text{ mA} \\ = 3 \mu\text{A}$$

Since active, $I_C = \beta I_B = 100 \times 0.003 = 0.3 \text{ mA}$

from the circuit,

$$I_C = \frac{10 - V_C}{R_C}$$

$$\Rightarrow V_C = 10 - R_C I_C = 10 - 3 \times 0.3$$

$$\Rightarrow \boxed{V_O = V_C = 9.1 \text{ V}}$$

VERIFICATION: Since we assumed **ACTIVE**, we

need to check if $V_{CE} > 0.2 \text{ V}$

Since $V_E = 0 \Rightarrow V_{CE} = V_C - V_E = 9.1 \text{ V} \Rightarrow \underline{\underline{\text{Assumption Correct}}}$
(yay!)

② Assume BJT in **ACTIVE**

$$\Rightarrow I_B = \frac{V_i - V_B}{R_B} = \frac{5 - 0.7}{100} = 0.043 \text{ mA}$$

$$\Rightarrow I_C = \beta I_B = 4.3 \text{ mA}$$

$$\Rightarrow V_O = V_C = 10 - I_C R_C = 10 - 3 \times 4.3 = -2.9 \text{ V}$$

VERIFICATION: $V_{CE} = -2.9V \therefore V_{CE} < 0.2V$

$\Rightarrow$ Assumption WRONG! ☹️

Now assume in **SATURATION**

$$\therefore V_{CE} = 0.2V$$

Since $V_E = 0V$ and $V_{CE} = V_C - V_E$

$$\therefore V_C = V_o = 0.2V.$$

$$I_B = \frac{V_i - V_B}{R_B} = \frac{5 - 0.8}{100} = 0.043mA$$

$\rightarrow$ Because SAT $\Rightarrow V_{BE} = 0.8V$

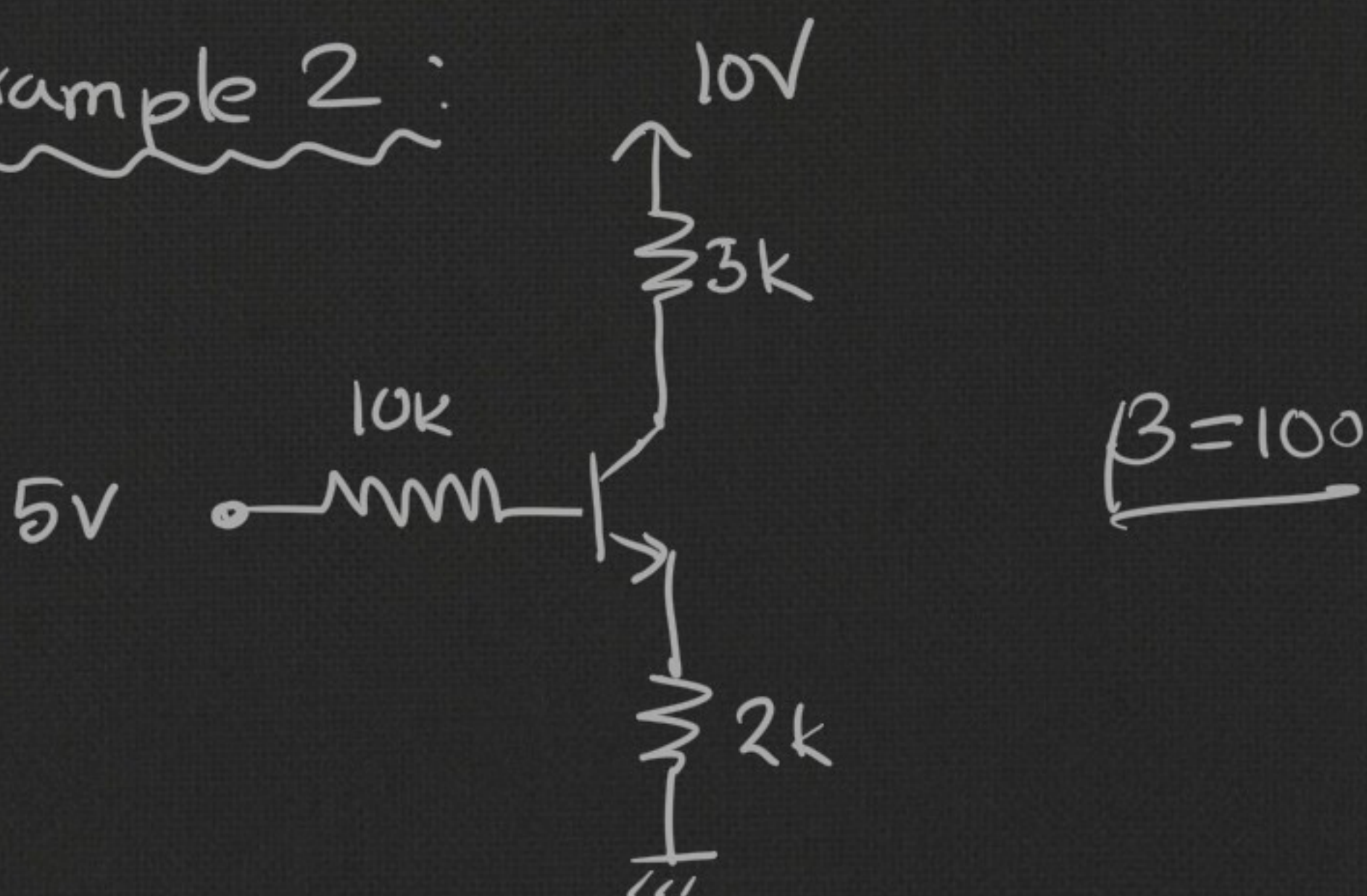
$$I_C = \frac{10 - V_C}{R_C} = \frac{10 - 0.2}{3} = 3.27mA$$

VERIFICATION: Since we've assumed saturation,

we need to show $\frac{I_C}{I_B} < \beta$. Now, $\frac{I_C}{I_B} = \frac{3.27}{0.043} = 76$

$\therefore \frac{I_C}{I_B} < 100 \Rightarrow$ Assumption correct. 😊

Example 2:



Find I_B , I_C , I_E and V_{CE} .

Solⁿ: Assume in **ACTIVE**

$$\Rightarrow I_C = \alpha I_E \text{ --- (i)}$$

$$I_C = \beta I_B \text{ --- (ii)}$$

from (i) and (ii), $\beta I_B = \alpha I_E$

$$\Rightarrow I_E = \frac{\beta}{\alpha} I_B \text{ --- (iii)}$$

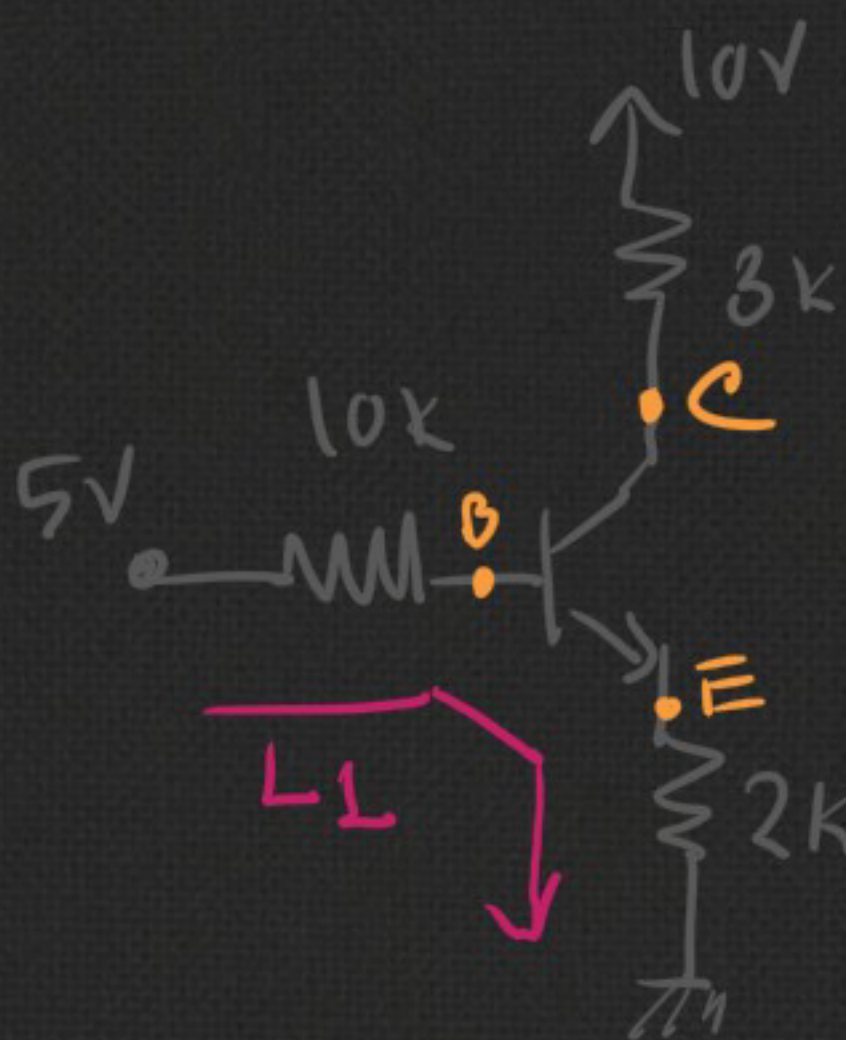
Writing KVL along $L_1 \Rightarrow$

$$5 = 10 I_B + V_{BE} + 2 I_E$$

$$\Rightarrow 5 = 10 I_B + 0.7 + 2 \times \frac{\beta}{\alpha} I_B \quad \rightarrow \text{from (iii)}$$

$$\Rightarrow \boxed{I_B = 0.02 \text{ mA}}$$

Since assumed
ACTIVE, $V_{BE} = 0.7$



$$\therefore I_C = \beta I_B = 100 \times 0.02 = 2 \text{ mA}$$

$$\therefore \boxed{I_C = 2 \text{ mA}}$$

$$\therefore I_E = I_C + I_B \Rightarrow \boxed{I_E = 2.02 \text{ mA}}$$

From the ckt, $V_E - 0 = I_E \times 2$

$$\Rightarrow V_E = 4.04 \text{ V}$$

Again, from the ckt, $10 - V_C = I_C \times 3$

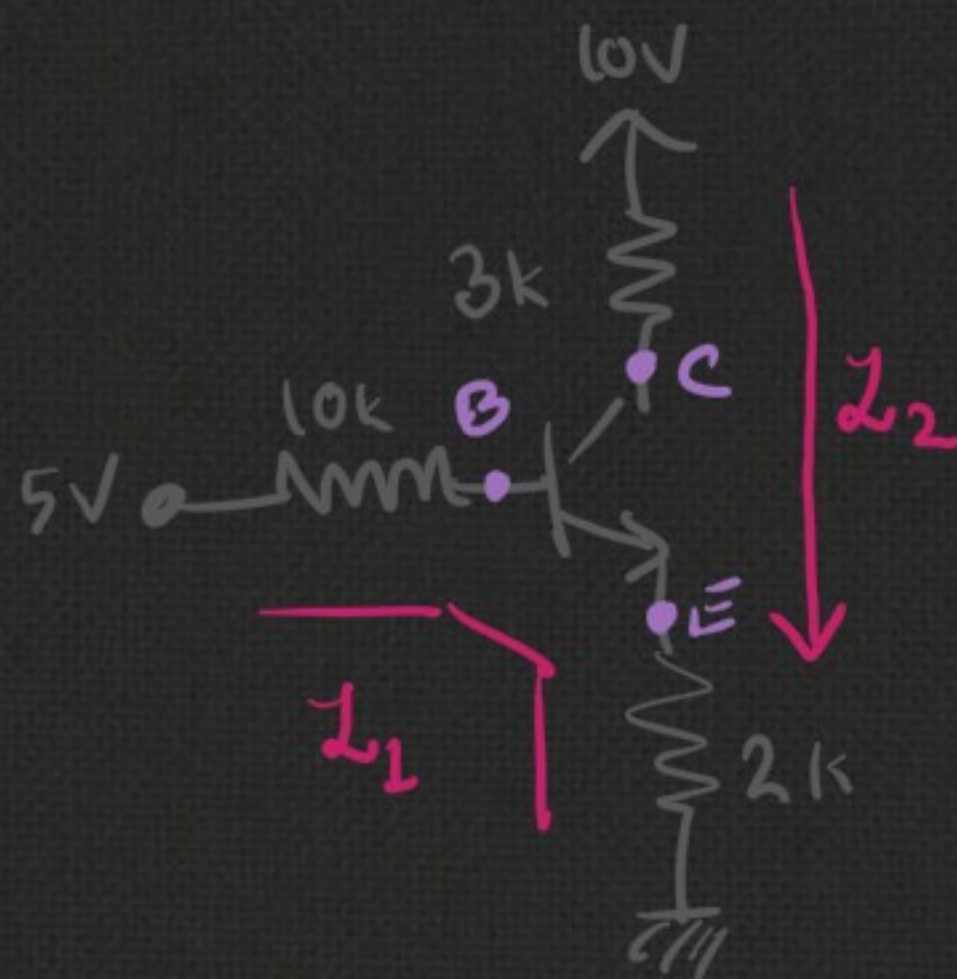
$$\Rightarrow V_C = 4 \text{ V}$$

$$\therefore V_{CE} = V_C - V_E$$

$$\Rightarrow \boxed{V_{CE} = 0.04 \text{ V}}$$

Assumed ACTIVE, found $V_{CE} < 0.2 \text{ V}$. So assumption WRONG! (- -)

Assume in SATURATION $\Rightarrow \boxed{\therefore V_{CE} = 0.2 \text{ V}}$



KVL along L_1

$$5 = 10I_B + V_{BE} + 2I_E$$

$$\Rightarrow 5 = 10I_B + 0.8 + 2I_E \quad \text{--- (1)}$$

KVL along L_2

$$10 = 3I_C + V_{CE} + 2I_E$$

$$\Rightarrow 10 = 3I_C + 0.2 + 2I_E \quad \text{--- (2)}$$

PTO

From KCL $\Rightarrow$

$$I_E = I_C + I_B \quad \text{--- (iii)}$$

Solving (i), (ii) and (iii)

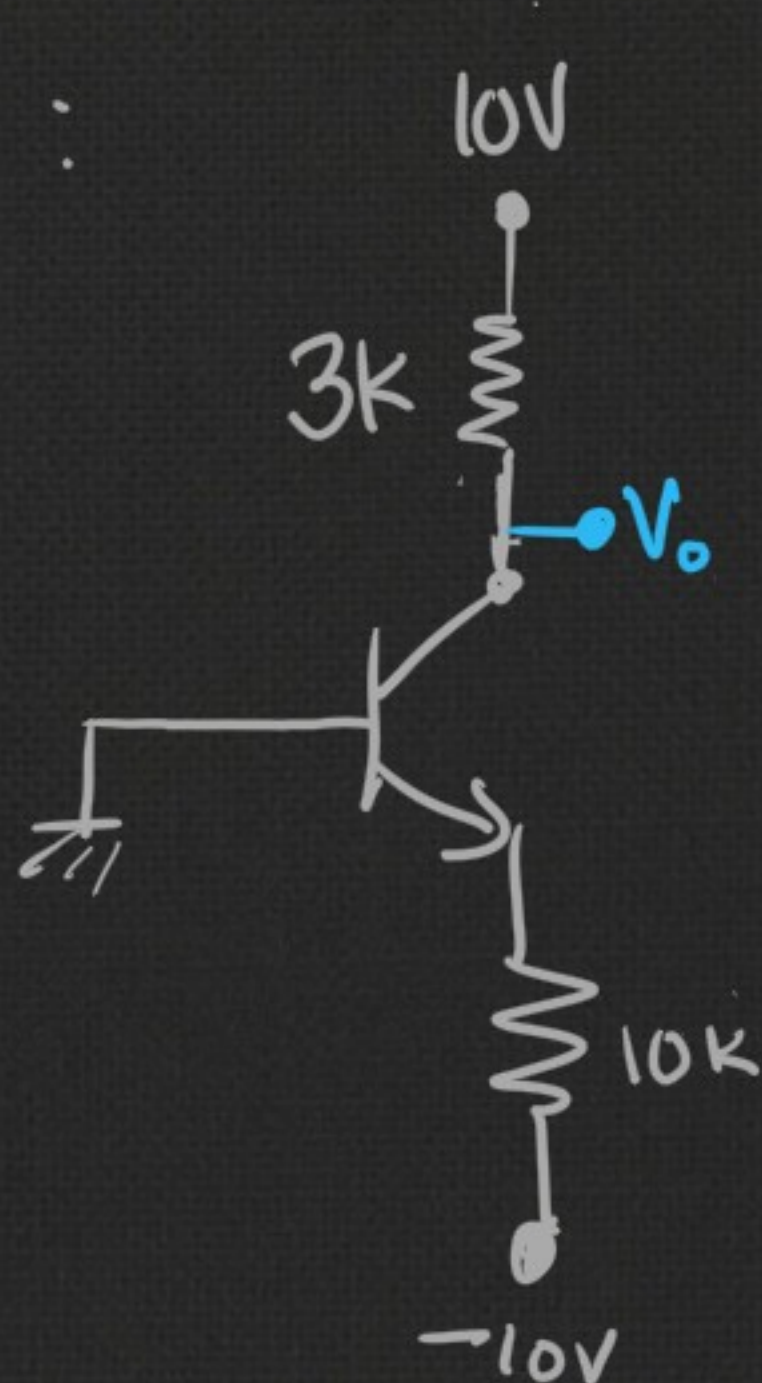
$$I_B = 0.03 \text{ mA}$$

$$I_C = 1.95 \text{ mA}$$

$$I_E = 1.98 \text{ mA}$$

Since $\frac{I_C}{I_B} = \frac{1.95}{0.03} < \beta$, so assumption CORRECT!

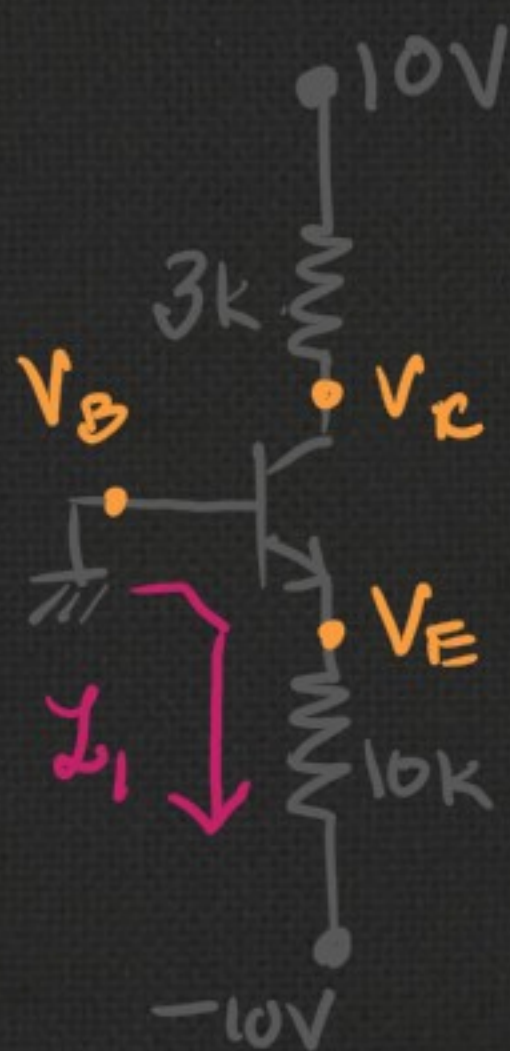
Example 3:



$$\beta = 100$$

find I_B , I_C , I_E and V_o

Soln:



Base connected to GND

$$\therefore V_B = 0V$$

Assume BJT **ACTIVE**

$$\therefore V_{BE} = 0.7V$$

$$\Rightarrow V_B - V_E = 0.7V$$

$$\Rightarrow V_E = -0.7V$$

KVL along L_2 : $0 = V_{BE} + 10I_E + (-10)$

$$\Rightarrow 0 = 0.7 + 10I_E - 10$$

$$\Rightarrow \underline{I_E = 0.93 \text{ mA}}$$

$$\therefore I_C = \alpha I_E = 0.99 \times 0.93 \Rightarrow \underline{I_C = 0.92 \text{ mA}}$$

$$\therefore I_B = I_E - I_C \Rightarrow \underline{I_B = 0.01 \text{ mA}}$$

$$\beta = 100$$

$$\therefore \alpha = \frac{1+\beta}{\beta} = 0.99$$

$$\text{From ckt, } 10 - V_C = 3I_C$$

$$\Rightarrow \underline{\underline{V_C = 7.24 \text{ V}}}$$

$$\text{Check: } V_{CE} = V_C - V_E = 7.24 - (-0.2)$$

$$\Rightarrow V_{CE} = 7.44 \text{ V}$$

$$\therefore V_{CE} > 0.2 \Rightarrow \underline{\underline{\text{Assumption correct!}}}$$